

Balancing Data Utility and Privacy in Medical Synthetic Data Generation: A Comparative Study of GAN-based and Large Language Model Approaches

Proposal for a Master Thesis

submitted by

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Contents

1	Introduction	3
2	Motivation	3
3	Research Questions	4
4	Literature Review	4
5	Methodology	5
6	Timeline	6
7	Conclusion	7

1 Introduction

The exponential growth of digital health records, coupled with advances in artificial intelligence (AI), has created an unprecedented opportunity to transform healthcare delivery. However, this potential is tempered by strict regulatory frameworks such as the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA), which limit access to sensitive medical data. Conventional anonymisation methods have repeatedly proven insufficient to guarantee privacy in high-dimensional health data, as re-identification and linkage attacks remain viable [1]. Synthetic data generation (SDG) has emerged as a promising alternative, enabling the creation of artificial datasets that retain the statistical properties of the original while mitigating privacy risks. Among the most prominent approaches are *Generative Adversarial Networks* (GANs) and *Large Language Models* (LLMs). GANs have demonstrated success in structured medical datasets and imaging data, while LLMs are particularly transformer-based models which offer advantages in modelling heterogeneous and unstructured records [2]. This thesis proposes a systematic, experimental comparison of GAN and LLM based SDG for healthcare, evaluating the trade-off between **data utility** and **privacy preservation** using unified metrics.

2 Motivation

The need for privacy-preserving data generation in healthcare is not only a technical challenge but a very important factor of medical research. Due to HIPAA and GDPR constraints on medical data sharing, data anonymization is an important enabler for research. The scarcity of openly accessible, high-quality clinical data constrains innovation in diagnosis, treatment planning, and epidemiological modelling. Regulatory restrictions and ethical imperatives make direct sharing of identifiable records infeasible. Synthetic data presents a pathway to alleviate this bottleneck by allowing the distribution of data without direct disclosure of personal identifiers, thus accelerating AI model development and validation. However, its adoption hinges on balancing two competing objectives:

1. **Utility:** The synthetic dataset must preserve the statistical and structural characteristics necessary for downstream analytical tasks.
2. **Privacy:** The dataset must provide strong resistance to adversarial attacks, such as membership inference and attribute disclosure.

While GANs and LLMs each offer unique strengths, there is limited comparative work evaluating their effectiveness under a unified privacy–utility framework, particularly for heterogeneous healthcare datasets.

3 Research Questions

1. Which synthetic data generation approach, based on GANs or Large Language Models, offers superior performance in balancing data utility and privacy preservation in medical datasets?
2. What are the privacy evaluation matrices and what are the trade-offs in utility and privacy in popular Synthesized Data Generation Models?

4 Literature Review

Due to the growing data generation, different data anonymization techniques has been developed for privacy reason. Techniques like generalization, suppression, distortion, swapping masking has been introduced.[3] But most of the techniques has one big drawback, utility preservation. Stadler et al. said while synthetic data often outperforms naive anonymization methods, achieving a stable trade-off between preserving meaningful information and safeguarding individual privacy remains difficult [4]. Zhang et al. [5], in their work on PrivSyn, embedded DP into the synthesis of high dimensional tabular data, ensuring that nuanced attribute correlations are maintained without disclosing sensitive records. SafeSynthDP demonstrates that LLM driven synthesis can be paired with differential privacy (e.g., Laplace/Gaussian mechanisms) to trace explicit ϵ -utility frontiers while maintaining competitive downstream accuracy and resisting membership inference [6]. They have adjusted hyperparameter for privacy and accuracy for utility metrics.

Miletic et al. [2] used a Framework called Tabula that treats each row as text and shows that Pythia, a family LLMs can match or surpass a strong tabular GAN (e.g., CTGAN) on TSTR utility, with gains increasing by model size [2]. The study focuses on *utility* rather than privacy. So in my research I will add privacy evaluations. ADS-GAN proposes a utility first, *non-DP* GAN that adds an *identifiability* regularizer: a weighted distance discourages generators from producing near duplicates of real patients, with rare features receiving higher penalties [7]. The paper reports strong realism (marginals/joint divergences) and competitive TSTR performance relative to baselines, while providing an interpretable privacy knob distinct from formal DP. It shows excellent results in both utility and privacy metrics.

I include HealthGAN as my non-DP GAN baseline because it was designed specifically for mixed-type EHR tables, comes with a complete secure training to model export workflow, and introduces nearest-neighbor Adversarial Accuracy (AA) to co-measure resemblance and privacy (TrainAA/TestAA and privacy-loss), yielding competitive TSTR utility while keeping a small, attributes that align with my governance and benchmarking needs. Murtaza et al. [1] demonstrated that it has been widely used for sythetic data generation [8][9] [10] which reinforces its status as a widely recognized baseline I can compare against DP GANs and LLMs in my study.

I select PATE-GAN as the differentially private generator because it enforces formal (ϵ, δ) -DP by decoupling privacy from adversarial training: multiple teacher discriminators are trained on disjoint data partitions, their votes are *noisily aggregated* to ensure DP, and a *student* discriminator learns from these noisy labels so the generator can be optimized end-to-end while inheriting DP by post-processing [11]. This design avoids exposing private gradients directly (unlike DPGAN) and provides a tunable privacy budget with an explicit accountant, which aligns with survey guidance to report *both* formal guarantees and empirical privacy audits [12, 1]. In domains where non-DP synthetic data has been shown to leave outliers vulnerable

[4], and where DP graphical approaches (e.g., PrivSyn) trade utility for stronger privacy [5], PATE-GAN offers a principled middle ground: a GAN-based model with controllable ϵ that typically delivers stronger task utility than DP-SGD at comparable privacy while supporting mixed-type tabular generation which making it a suitable DP baseline against which I will compare Health-GAN (non-DP) and LLM-based synthesis. Hernandez et al. [12] in his systematic review of tabular health data for synthetic data generation stated that no single method or metrics is universally best and listed out resemblance, privacy and utility metrics used by other authors. I took inspiration from the paper and accumulated all the evaluation metrics to do my evaluation on the chosen models and to find out if any standardization is possible.

5 Methodology

To answer RQ1 and RQ2, we will prepare the datasets with data pre-processing, next we will be implementing the models that are mentioned below and run utility and privacy metrics. We will try to determine if a standardized evaluation metrics is achievable.

5.1 Proposed Dataset

1. **UCI Diabetes 130-US Hospitals:** The Diabetes dataset offers a large, mixed-type, encounter-level table with class imbalance and many categorical fields, making it a useful stress test for synthesis realism and downstream classification utility [13]. I use it as a complementary benchmark to MIMIC-III for reproducible, public experiments and fine-grained metric comparisons [13].
2. **MIMIC-III:** MIMIC-III is a widely used, de identified critical-care resource with rich mixed-type variables and outcomes suitable for realistic tabular synthesis and TSTR evaluations [14]. I use it as a primary benchmark to test scalability, mixed-type handling, and privacy auditing under clinical relevance [14].

5.2 Proposed Models

1. **Health-GAN (non-DP)**
2. **PATE-GAN (DP)**
3. **Pythia LLM**

5.3 Evaluation Metrics

5.3.1 Utility Metrics

1. Predictive model accuracy.
2. Kolmogorov–Smirnov (KS) statistic,
3. correlation matrix similarity,

4. Feature-wise distribution matching,
5. Joint-distribution similarity,
6. AUROC

5.3.2 Privacy Metrics

1. Membership Inference Attack (MIA) success rate.
2. Attribute inference.
3. Distance-to-Closest Record (DCR)
4. Clustering / proximity as a proxy (DBSCAN)
5. (ϵ)-identifiability
6. Distributional closeness metrics: Wasserstein Distance, k-anonymity.

5.4 Training and Generation

1. Data acquisition and understanding to gain domain specific knowledge
2. Preprocess datasets according to model-specific requirements (e.g., mode-specific normalization for GANs, tokenisation for LLMs).
3. Train models with equal computational budgets and comparable data splits.
4. Generate fully synthetic datasets from each model for evaluation.

6 Timeline

Month	Task
September/2025	Literature review finalisation, dataset acquisition.
October/2025 - December/2025	Preprocessing, GAN model implementation.
November/2025 - January/2026	LLM model implementation.
December/2025 - January/2026	Evaluation and standardization experiments
December/2025 - February/2026	Thesis write-up, revision, and submission.

Table 1: Six-month project timeline.

7 Conclusion

In this thesis, I will conduct a controlled comparison of GAN and LLM-based methods for privacy-preserving synthetic medical data under a single, transparent evaluation protocol. Using mixed-type clinical benchmarks (MIMIC-III and Diabetes 130-US Hospitals) and representative models (Health-GAN as a non-DP baseline, PATE-GAN with formal (ϵ, δ) -DP, and a Pythia-family LLM), the study will quantify both utility and privacy trade-offs with aligned training budgets and standardized audits. By pairing task-based utility (e.g. TSTR performance, feature and joint-distribution similarity) with privacy risk assessments (e.g., membership and attribute inference, distance-to-closest record, clustering proximity), the work aims to identify where each paradigm is preferable and why.

Beyond reporting comparative results, the thesis contributes a reusable evaluation workflow and practical guidance for healthcare data stewards: when a non-DP GAN suffices, when formal DP via PATE-GAN is warranted, and where LLM-based synthesis is competitive given privacy budgets and governance constraints. While the scope is limited to tabular EHR-like data and a focused model set, the protocol lays groundwork for extending to longitudinal settings and stronger DP mechanisms for LLMs. Ultimately, the findings are intended to inform method selection and support emerging standards for safe, high utility data sharing in healthcare.

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